Recommendation Systems Datasets

CP4285 Modern Recommendation Systems — Auxiliary project resource

CP4285 Instruction Team

15 Sep 2026

Scan for the slides
soc-n.us/cp4285-t2610-datasets

01 Overview

CHANGELOG

- **15 Sep 2026** — Created the project-planning guide, expanded the catalogue to the full source inventory, and added a deck QR link.

What this deck is for

- Find a **public, reproducible** dataset that makes a recommendation question possible.
- Match the data modality to the claim: ratings, sessions, text, social links, location, or impressions.
- Start small enough to establish classical and neural baselines before attempting a novel model.

> ⚠ **Warning**
>
> Dataset scale is not project quality. A well-scoped question, credible split, error analysis, and ethical critique matter more than a large download.

How to read a dataset slide

RECORDS
Published count
Interactions, impressions, or check-ins

FOOTPRINT
Download / RAM
Published where available; otherwise an estimate

FIT
Task + risk
The research claim the data can support

- **Published download** is the provider’s compressed archive size when stated.
- **Planning RAM** is an indicative laptop/cluster budget for a parsed working subset; it varies with filtering, features, framework, and batch size.
- Every team must check the current licence, documentation, and terms before download.



02 Dataset Catalogue

Catalogue key

The catalogue is a **source inventory**, not a recommendation that every dataset is equally suitable or still equally accessible.

- ★ has a full project-planning slide in Section 03.
- Entries marked **textbook** were identified in the course books but are not separate RecBole entries.
- Dataset availability, licences, versions, and download sizes change; check the linked provider before building.

Catalogue: shopping and transaction data I

Dataset family / version	Primary signal	Detail
Amazon Reviews 2014	Reviews + product metadata	24 categories; 142.8M reviews
Amazon Reviews 2018	Reviews + product metadata	29 categories; 233.1M reviews
★ Amazon Reviews 2023	Reviews, metadata, images	44 categories; 571.5M reviews
Amazon-M2	Multilingual customer sessions	Six locales
Alibaba-iFashion	Outfit composition + item attributes	Fashion compatibility

Catalogue: shopping and transaction data II

Dataset family / version	Primary signal	Detail
★ Epinions	Ratings + directed trust links	Social/trust recommendation
Yelp 2018 / 2020 / 2021 / 2022 / full	Reviews, businesses, tips, check-ins	Versioned business-review releases
★ Yelp Open Dataset	Reviews + business attributes	Current scoped project option
Tmall	E-commerce interactions	IJCAI-16 competition data
Ta Feng	Grocery transactions	Nov 2000–Feb 2001

Catalogue: sessions and e-commerce logs

Dataset	Primary signal	Detail
★ Diginetica	Search sessions, clicks, purchases	Queries, product metadata, prices
★ YOOCHOOSE / RecSys Challenge 2015	Clickstream sessions + purchases	Sequential ranking benchmark
★ Retailrocket	Views, carts, transactions	Raw hashed events
Tmall sessions (textbook)	E-commerce sessions	1.77M sessions in the course book table
XING (textbook)	Job-search / interaction logs	Sequential/industrial recommendation

Catalogue: advertising and CTR prediction

Dataset	Primary signal	Detail
Criteo	Ad impressions + click labels	Several days of traffic
Avazu	Ad click-through logs	CTR prediction contest
iPinYou	Real-time bidding logs	Three competition seasons
AliEC	Taobao display-ad clicks	CTR prediction

Catalogue: places and spatiotemporal data

Dataset	Primary signal	Detail
Foursquare	Check-ins, time, GPS, venue semantics	NYC and Tokyo collection
★ Gowalla	Check-ins + social graph	6.44M check-ins

Catalogue: movies, video, and streaming

Dataset	Primary signal	Detail
★ MovieLens 1M	Explicit movie ratings	Detailed project slide; MovieLens family
MovieLens 100K / 30M (textbook)	Sessionised rating events	Course-book benchmark variants
Netflix Prize	Movie ratings	Official competition data
Douban Movie	Movie comments	Chinese social-media context
Twitch-100K / Twitch-full	Streams + chat connections	43-day collection

Catalogue: music and audio

Dataset	Primary signal	Detail
★ Last.fm 2K	Listening, tags, friendships	HetRec 2011 release
LFM-1b	Timestamped listening events	1B+ events; 120K+ users
Yahoo Music	Music-community preferences	Artist preference snapshot
KGRec-music / KGRec-sound	Implicit feedback, text, tags	Knowledge-graph-oriented data
Music4All-Onion	Music audio/video/metadata	109,269 music pieces

Catalogue: books, games, images, and jokes

Dataset	Primary signal	Detail
Book-Crossing	Explicit + implicit book ratings	Demographics included
GoodReads	Shelves, ratings, reviews	Multiple interaction levels
Steam	Reviews, play hours, game metadata	7.79M reviews
Anime	User ratings	MyAnimeList preferences
Pinterest	Image/social recommendation	Content-based image recommendation
Jester	Anonymous joke ratings	Jester recommender data

Catalogue: learning and health data

Dataset	Primary signal	Detail
KDD Cup 2010	Student exercise attempts	Educational data mining
EndoMondo	Workout and sensor logs	Health-sensitive sequential data

Catalogue: web, news, reviews, and apparel

Dataset	Primary signal	Detail
Phishing Websites	Website features + labels	Classification, not user-item recommendation
Behance	Likes + image data	Anonymised art-community subset
★ MIND-small	News impressions + article text	Detail slide uses a manageable subset
BeerAdvocate / RateBeer	Multi-aspect beer reviews	Taste, look, feel, smell
ModCloth / RentTheRunway	Clothing-fit feedback	Fit measurements
Adult	Census attributes + income label	Classification; not a recommender dataset

Catalogue: course-text datasets beyond RecBole

Dataset	Primary signal	Detail
★ Adressa	News reading sessions + articles	16-day processed benchmark
Reddit	Community / session interactions	1.14M sessions in course-book table
★ CiteULike-a	User-paper saves + tags	Scholarly recommendation
BBC News (LLM book)	News articles + text	Text-based recommendation example
Amazon Fashion (LLM book)	Reviews, images, product text	A scoped Amazon 2023 category

03 Selected Dataset Details ⭐

MovieLens 1M

INTERACTIONS
1,000,209
6,040 users · 3,706 movies

FOOTPRINT
6 MB ZIP
Under 1 GB RAM for standard experiments

DATA
Ratings
1-5 stars, timestamps, genres, demographics

Best fit: transparent CF or ranking baselines; factorisation, neural CF, and genre-aware hybrids.

Papers: Harper & Konstan (2015), *The MovieLens Datasets*; Koren, Bell & Volinsky (2009), *Matrix Factorization Techniques for Recommender Systems*.

URL: https://grouplens.org/datasets/movielens/1m/

> ⚠ **Warning**
>
> The demographic fields are coarse and historic. Do not claim they represent contemporary audiences or use them casually as protected-attribute ground truth.

Amazon Reviews 2023 - select one category

CATALOGUE SCALE
571.5M
Reviews across 44 categories

FOOTPRINT
Varies
Use one category + a documented k-core

DATA
Hybrid
Ratings, reviews, metadata, images by category

Best fit: item cold start, review-aware recommendation, retrieval from product text, or an LLM-assisted explanation study.

Papers: McAuley et al. (2015), *Image-based Recommendations*; Hou et al. (2024), *Amazon Reviews 2023*.

URL: https://amazon-reviews-2023.github.io/

> ⚠ **Warning**
>
> Choose a category before downloading. Reviews are not neutral preference labels; helpfulness, popularity, and seller incentives can bias what is observed.

Yelp Open Dataset

REVIEWS
6.99M
150,346 businesses · 11 metro areas

FOOTPRINT
~4.4 GB ZIP
Plan 8-16 GB RAM for full JSON processing

DATA
Reviews + POIs
Text, stars, attributes, hours, check-ins

Best fit: business/POI recommendation using text and metadata; city-level temporal split; exposure and representation analysis.

Source / papers: Yelp (2022), *Yelp Open Dataset*; He et al. (2017), *Neural Collaborative Filtering* (a useful rating-ranking baseline).

URL: https://business.yelp.com/data/resources/open-dataset/

MIND-small

BEHAVIOUR LOGS
230,117
94,057 users · 65,238 news articles

CLICKS
347,727
Use provided train / dev divisions

DATA
Impressions + text
Title, abstract, category, entities, click history

Best fit: candidate ranking, news encoders, sequential interests, and ranking metrics such as AUC, MRR, and nDCG.

Papers: Wu et al. (2020), *MIND*; Wu et al. (2019), *NAML: Neural News Recommendation with Attentive Multi-View Learning*.

URL: https://msnews.github.io/

> ⚠ **Warning**
>
> Non-clicked impressions are observed non-clicks, not certain dislikes. Discuss position/exposure bias, short news life cycles, and misinformation harms.

Diginetica

SESSIONS
204,789
993,483 click events

ITEMS
43,136
Average session length: 4.85

DATA
Search sessions
Clicks, queries, product metadata, prices

Best fit: next-item ranking on short anonymous sessions; compare item-kNN, GRU4Rec, and Transformer/attention models.

Papers: Kotsifakos et al. (2019), *Diginetica: A Dataset for Session-based Recommendation*; Hidasi et al. (2016), *Session-based Recommendations with Recurrent Neural Networks*.

URL: https://competitions.codalab.org/competitions/11161

YOOCHOOSE / RecSys Challenge 2015

SESSIONS
7.98M
31.71M click/buy events

ITEMS
37,486
182 days · mean length 3.97

DATA
Sessions
E-commerce clickstream and purchase signal

Best fit: an ambitious sequential project. Work on a chronologically defined subset before scaling.

Papers: Hidasi et al. (2016), *Session-based Recommendations with RNNs*; Tan et al. (2016), *Improved Recurrent Neural Networks for Session-based Recommendations*.

URL: https://2015.recsyschallenge.com/challenge.html

> ⚠ **Warning**
>
> This is a session dataset, not a long-term personalised-profile dataset. Avoid claims about stable user preference or demographic fairness that the data cannot support.

Retailrocket

EVENTS
2.76M
1.41M visitors · 235,061 items

FOOTPRINT
~340 MB CSV
Plan 2-4 GB RAM after parsing

DATA
Events
View, add-to-cart, transaction, timestamps

Best fit: event-type-aware sequential recommendation; compare whether carts/purchases should be weighted differently from views.

Papers: Hidasi & Karatzoglou (2018), *Recurrent Neural Networks with Top-k Gains*; Quadrana, Cremonesi & Jannach (2017), *Sequence-Aware Recommender Systems*.

URL: https://www.kaggle.com/datasets/retailrocket/ecommerce-dataset

Adressa

SESSIONS
982,210
2.65M click events in a 16-day processed version

ITEMS
13,820
Short sessions: mean length 2.7

DATA
News sessions
Reading events, article metadata/content by release

Best fit: short-horizon news recommendation and temporal evaluation; compare session-only and text-aware ranking.

Papers: Gulla et al. (2017), *The Adressa Dataset for News Recommendation*; Karimi et al. (2018), *News Recommender Systems*.

URL: https://reclab.idi.ntnu.no/dataset/

> ⚠ **Warning**
>
> News preference is volatile and politically consequential. Preserve time order, avoid leaking future article metadata, and analyse publisher/topic concentration.

Last.fm 2K

INTERACTIONS
92,834
1,892 users · 17,632 artists

FOOTPRINT
Small ZIP
Laptop-friendly; under 1 GB RAM

DATA
Implicit + social
Listening counts, tags, user friendships

Best fit: tag-aware, social, and implicit-feedback recommendation; investigate diversity beyond head artists.

Papers: Cantador, Brusilovsky & Kuflik (2011), *2nd HetRec Workshop*; Kula (2015), *Metadata Embeddings for User and Item Cold-start Recommendations*.

URL: https://grouplens.org/datasets/hetrec-2011/

Gowalla

CHECK-INS
6.44M
196,591 users · Feb 2009-Oct 2010

SOCIAL GRAPH
950,327
Friendship edges; 107,092 users appear in check-ins

DATA
Spatiotemporal
Time, latitude, longitude, POI, friendship

Best fit: next-POI recommendation, geographical influence, or social/context-aware ranking.

Papers: Cho, Myers & Leskovec (2011), *Friendship and Mobility*; Ye, Yin & Lee (2011), *Exploiting Geographical Influence for Collaborative POI Recommendation*.

URL: https://snap.stanford.edu/data/loc-gowalla.html

> ⚠ **Warning**
>
> Precise location traces are sensitive. Minimise retained coordinates, avoid re-identification claims, and do not visualise individual mobility paths without a strong reason.

Epinions

RATINGS
664,824
49,290 users · 139,738 items

TRUST LINKS
487,183
Directed user-user edges

DATA
Ratings + trust
Historic consumer-review community data

Best fit: trust propagation, social regularisation, robustness, and cold-start comparison against a rating-only baseline.

Papers: Massa & Avesani (2006), *Trust-aware Recommender Systems*; Guo, Zhang & Yorke-Smith (2015), *TrustSVD*.

URL: https://www.cse.msu.edu/~tangjili/trust.html

> ⚠ **Warning**
>
> Trust links are neither ground-truth expertise nor similarity. The data are historical, so frame claims as benchmark evidence rather than contemporary behaviour.

CiteULike-a

USER-PAPER SAVES
204,986
5,551 users · 16,980 papers

FOOTPRINT
Small
Laptop-friendly after text/tag filtering

DATA
Implicit + tags
Paper saves, tags, and textual content by release

Best fit: scholarly-paper recommendation, collaborative topic models, tag-aware retrieval, or an LLM-based text encoder against a clean baseline.

Papers: Wang & Blei (2011), *Collaborative Topic Modeling for Recommending Scientific Articles*; Nallapati et al. (2008), *Joint Latent Topic Models for Text and Citations*.

URL: https://github.com/js05212/citeulike-a

04 Common Practices



Choose a dataset by the claim you can defend

1. Define the task

Who receives what recommendation, from which signal, at what time?

2. Lock the split

Use a chronological split for sequential, news, or context-sensitive claims.

3. Establish baselines

Start with popularity and a classical model before a neural/LLM system.

4. Evaluate the interface

Use ranking metrics for top-k recommendation; inspect coverage and diversity too.

5. Analyse failure

Segment by activity level, item popularity, time, and relevant content groups.

6. Address people

Document licence, provenance, privacy, exposure, representation, and likely harms.

A reproducibility checklist

- Record the exact source URL, download date, licence/terms, version, and checksum where supplied.
- Define every filter: event types retained, k-core threshold, sessionisation rule, and minimum sequence length.
- Split before fitting any data-dependent representation; do not let future interactions or item text leak into training.
- Publish a configuration for popularity, ItemKNN/BPR/MF, and the advanced model on the same split.
- Report resource use: sampled records, download size, GPU/CPU, RAM, training time, and random seeds.
- Separate an observed non-click from a negative preference unless the task design justifies the assumption.

